

```
ubuntu ansible_host=192.168.122.244 ansible_connection=ssh
ansible_user=ubuntu ansible_password=password
```

You should now be able to issue commands to the guest OS, using Ansible. For example:

```
$ ansible -i inventory/kvm/inventory ubuntu -m ping
ubuntu | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
```

Installing Nginx

The Nginx software package in Ubuntu can be installed on the guest machine. The APT package repository is first updated before installing the Nginx Web server. The Uncomplicated Firewall (UFW) is then used to enable both HTTP and HTTPS access on the guest OS. The Web server is then started, and the playbook waits for the server to listen on port 80. The Ansible playbook is provided below for reference:

```
---
- name: Install nginx
  hosts: ubuntu
  become: yes
  become_method: sudo
  gather_facts: true
  tags: [nginx]

tasks:
  - name: Update the software package repository
    apt:
      update_cache: yes

  - name: Install nginx
    package:
      name: "{{ item }}"
      state: latest
    with_items:
      - nginx

  - name: Allow Nginx Full
    ufw:
      rule: allow
      name: Nginx Full
      state: enabled

  - name: Allow Nginx Full
    ufw:
      rule: allow
      name: OpenSSH
      state: enabled

  - name: Start nginx
```

```
service:
  name: nginx
  state: started
```

```
- wait_for:
  port: 80
```

The above playbook can be invoked as follows:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/nginx-ssl.yml --tags nginx -K
```

The `-K` option prompts for the sudo password of the Ubuntu user. You can append multiple `-v` to the end of the playbook invocation to get a more verbose output.

If you open a browser on the host system with the URL `http://192.168.122.244`, you should see the default Nginx home page as shown in Figure 1.

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org. Commercial support is available at nginx.com.

Thank you for using nginx.

Figure 1: Nginx home page

Generating SSL certificates

The required SSL certificates need to be created using Ansible. The OpenSSL and Python-openssl packages are installed after updating the APT software repository in Ubuntu. An OpenSSL private key is generated in the `/etc/ssl/private/ansible.com.pem` file. The `/etc/ssl/csr` directory is created before generating the OpenSSL certificate signing request (CSR) with the required certificate parameters in the `/etc/ssl/csr/www.ansible.com.csr` file. The actual self-signed certificate is then generated as shown in the following playbook:

```
- name: Create SSL certificates
  hosts: ubuntu
  become: yes
  become_method: sudo
  gather_facts: true
  tags: [ssl]

tasks:
  - name: Update the software package repository
    apt:
      update_cache: yes

  - name: Install openssl
    package:
```